

Practice Quiz 07B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Use a limit hypothesis to obtain a local inequality.

1. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and $c \in \mathbf{R}$, and suppose $\lim_{x \rightarrow c} f(x) = 3$. Use the epsilon-delta definition to prove that there is a $\delta > 0$ such that, for every real x ,

$$0 < |x - c| < \delta \quad \longrightarrow \quad f(x) > 2.$$

Solution Apply the definition of $\lim_{x \rightarrow c} f(x) = 3$ with $\varepsilon = 1 > 0$. It gives a $\delta > 0$ such that every real x with $0 < |x - c| < \delta$ satisfies $|f(x) - 3| < 1$.

Let x be any real number with $0 < |x - c| < \delta$. Since $|f(x) - 3| < 1$, we have $-1 < f(x) - 3 < 1$, and hence $f(x) > 2$. The chosen δ therefore has the required property. ■